

B Pazhani Vel

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Motivated Computer Science student with a strong foundation in Machine Learning, Deep Learning, and MLOps. Experienced in building end-to-end ML pipelines — from data preprocessing to deployment. Passionate about solving real-world problems.

EDUCATION

Anna University - MIT Campus

July 2024 – present

Bachelor of Technology - Information Technology

CGPA : 9.7

Shanthinikethan Mat Hr Sec School

June 2022 – May 2024

10th Grade: 94% — 12th Grade: 91%

PROJECTS

Product Recommendation System

GitHub

Python | Surprise (SVD) | TF-IDF | Scikit-learn | Streamlit

- Engineered a hybrid recommendation engine combining SVD-based collaborative filtering and TF-IDF cosine similarity on Amazon Reviews dataset (500K+ interactions), blending scores with a weighted fusion model.
- Trained and evaluated SVD matrix factorization with hyperparameter tuning (**n_factors**, **reg**), achieving **RMSE of 0.91** and **Precision@10 of 74%** on a held-out test set.
- Deployed an interactive Streamlit app enabling real-time top-N product recommendations, covering **85%+** of the product catalog.

RAG-Based AI Assistant

GitHub

Python | LangChain | FAISS | Groq/OpenAI API | HuggingFace Embeddings

- Built a Retrieval-Augmented Generation pipeline using LangChain with a FAISS vector store and HuggingFace sentence embeddings, enabling document-grounded Q&A over custom knowledge bases.
- Reduced LLM hallucination rate by **~60%** compared to vanilla GPT prompting by grounding every response in retrieved context chunks with similarity threshold filtering.
- Achieved an **average query response time under 2 seconds** on 50+ page documents via chunk-based indexing and top-*k* retrieval (*k* = 5).

Student Score Predictor — MLOps on AWS

GitHub

Python | Scikit-learn | Flask/FastAPI | MLflow | AWS EC2

- Developed an end-to-end ML pipeline for student performance prediction with automated preprocessing, model training (Random Forest, XGBoost, Ridge), and MLflow experiment tracking across **15+ runs**.
- Deployed a REST API on AWS EC2 with Flask, serving real-time predictions with **model versioning and artifact registry** via MLflow, enabling one-click rollback to the best-performing model.
- Achieved an **R² score of 0.87** on test data; the MLOps setup reduced manual retraining effort by **~70%** through automated logging of metrics, parameters, and model artifacts.

TECHNICAL SKILLS

Languages: C ,C++ ,Python

Web Development: HTML ,CSS ,React.js ,Node.js ,Flask ,Bootstrap

Database: SQL ,MongoDB ,AWS

Core Concepts: DSA, DBMS, OOPS

Libraries: Pandas ,Numpy ,Matplotlib

HACKATHONS & COMPETITIONS

Barclays Hackathon — National Finalist

Barclays India

Alternate Credit Scoring System for MSMEs | Python | XGBoost | Random Forest | NLP

- Built a hybrid ML credit scoring engine for MSMEs using financial + alternative data (GST returns, digital footprint), enabling credit access for businesses with no formal credit history.
- Advanced to **national finals** out of hundreds of teams, finishing within **2 points** of the winning score — recognized for real-world impact and model interpretability.

CERTIFICATIONS

Pandas– Kaggle (2026)

Python for Data Science– NPTEL (2026)

C++ Programming– Udemy (2026)

Software Engineering Job Simulation – Forage (2026)

ACHIEVEMENTS

Secured **1st place** in a college-level Coding Contest – (2025)